Oluwadamilola (Dammy) Owolabi

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SUMMARY

Electrical engineer with an MS in Data Science who designs medium- and low-voltage distribution systems for investor-owned utilities, pairing load calculations, short circuit analysis, and duct bank design with Python, SQL, R, and Power BI to turn grid and operational data into decisions.

EXPERIENCE

Associate Distribution Engineer, Leidos — Houston, TX | Sep 2025 to Present

- Performed load calculations, short circuit analysis, and conduit sizing and fill checks for medium- and low-voltage distribution designs for CenterPoint Energy and Duke Energy.
- Sized transformers, step banks, conductors, fuses, and poles, drafted design schematics, and engineered underground duct bank layouts, including site walkdowns.
- Built material lists for designs and SAP work orders and ran estimate-versus-actual cost and material variance checks to identify root causes.
- Performed QA/QC on design packages, mentored newer designers, built a reusable team design template, and tracked team work in Jira dashboards.

Electrical Estimator, Ferreira Co. (PGE) — Irvine, CA | Jun 2023 to Apr 2024

- Estimated material and labor costs for electrical distribution and infrastructure projects using estimating software, pole simulations, and AutoCAD.
- Designed electrical service for residential solar and battery storage customers.
- Cut material costs by a few hundred dollars per project, about 10%, by removing unnecessary materials and labor from estimates.
- Created SAP work orders and notifications, built material lists, and ran estimate-versus-actual variance checks.

System IP Intern, Arm Ltd. — Austin, TX | Jun 2021 to Aug 2021

- Wrote Python scripts with fellow interns to automate collection and analysis of simulation results using Linux, Excel, and SQL.
- Reduced weekly simulation processing time from about 20 hours to about 12 hours.
- Diagnosed and resolved crashed simulation runs, reducing crashes from about 20 per week to about 2.
- Analyzed System IP performance metrics in Python and Excel and presented results to stakeholders.

Research Intern, Rice University Polymer Engineering Laboratory — Houston, TX | May 2019 to Aug 2019

- Fabricated and tested active-layer films for organic photovoltaics, collecting over 250 data points.
- Performed statistical analysis of experimental results in Excel and Python.
- Presented weekly progress and findings to the principal investigator.
- Co-authored a peer-reviewed paper published in Advanced Optical Materials (2020).

EDUCATION

M.S., Data Science, Southern Methodist University — Dallas, TX | Aug 2025

B.S., Electrical Engineering, Texas A&M University — College Station, TX | May 2022

Relevant coursework: load flow and fault studies, power electronics, statistical modeling, data visualization, databases.

PROJECTS

AI-Driven Optimization of Wind Energy Distribution in Texas (SMU capstone, first author, paper in draft): Built a Python multi-agent reinforcement learning environment with wind, load, and flywheel storage agents trained with Deep Q-Networks on 2.7M+ ERCOT demand and wind generation records (2014 to 2025) to reduce wind curtailment and transmission congestion.

Cotton Drone Analysis (Texas A&M senior design, project lead): Built a drone-based crop data system with Python, Django, Raspberry Pi, and Google Cloud.

PUBLICATION

Hu, Z., Sun, C., Lin, A., Jackson, J., Terlier, T., **Owolabi, D.**, et al. Improved Mechanical Durability of High-Performance OPVs Using Semi-Interpenetrating Networks. *Advanced Optical Materials*, 8(18), 2000516 (2020).

SKILLS AND CERTIFICATIONS

Programming and data: Python, SQL, R, C++, Excel (pivot tables, XLOOKUP, Power Query, VBA)
Analytics and visualization: Power BI, Tableau, D3.js, statistical modeling, machine learning
Power systems: load calculations, short circuit analysis, duct bank design, PowerWorld, PSS/E, PSCAD, MATLAB/Simulink
Tools: AutoCAD, GIS/ESRI, SAP, Jira, Git/GitHub, AWS (AWS Certified Cloud Practitioner)